Simplified explanation

From the above we can break the mechanism of the lean-in transient down into a few simple actions as follows:

- The initial counter-steering produces a slip angle which gives a tyre force out from the turn.
- This force produces a roll motion into the corner. Fig. 4.20 & 4.21.
- As the lean builds, gravity augments the process and the roll velocity and roll angle increase.
- The tyres react to the lean or camber angle by producing inward camber forces, which tend to slow the lean towards the final angle as well as pushing the bike into the turn. Rider steering input will modify these tyre forces until the desired cornering state is reached.
- Gyroscopic reactions play an in-significant part in adding to the total roll torque.
- The roll velocity gives rise to relatively large gyroscopic steering (not roll) torques which act against the rider's input and are an essential negative feedback to stabilize the whole system.

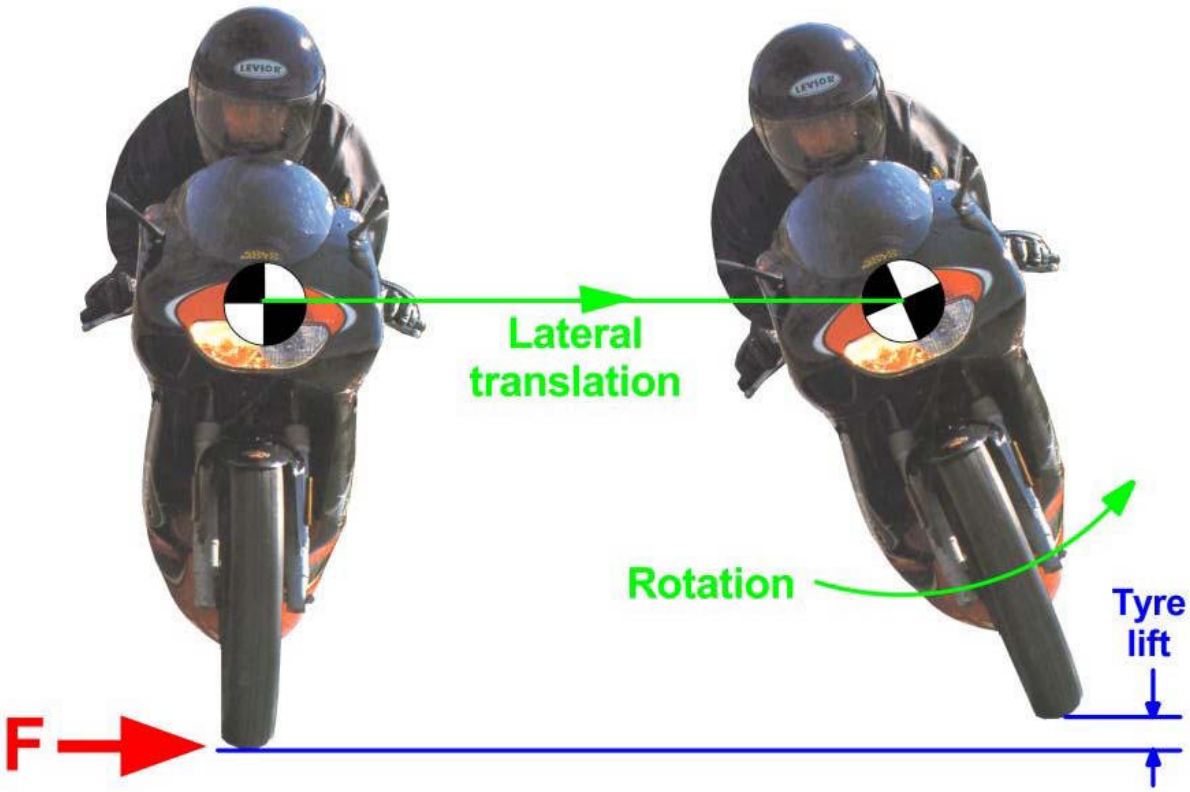

Fig. 4.20 **Counter-steering causes a force at the road/tyre junction outwards from the desired turn. This force has two effects. It causes a lateral translation of the CoG, shown exaggerated here. It also causes a rotation of the bike and rider into the corner. Once some lean angle and roll velocity has been generated by this method, gravity will take over to finish the lean-in. It can be seen that the rotation about the CoG tends to lift the tyre, momentarily reducing the vertical tyre force, hence reducing the maximum lateral force possible.**

We have so far considered the tyre force purely from the aspect of creating a torque about a horizontal roll axis, unfortunately it is more complex than that. Counter-steering produces a force at the front tyre, initially, until a yaw or roll angle is created, the rear tyre produces no lateral force. Reference to fig. 4.21 shows that the point of application of the front tyre force is both below and in front of the CoG. Therefore, both roll and yaw moments are produced, and so the resulting rotation is a combination of roll and yaw. The rotation axis will point upwards towards the front.

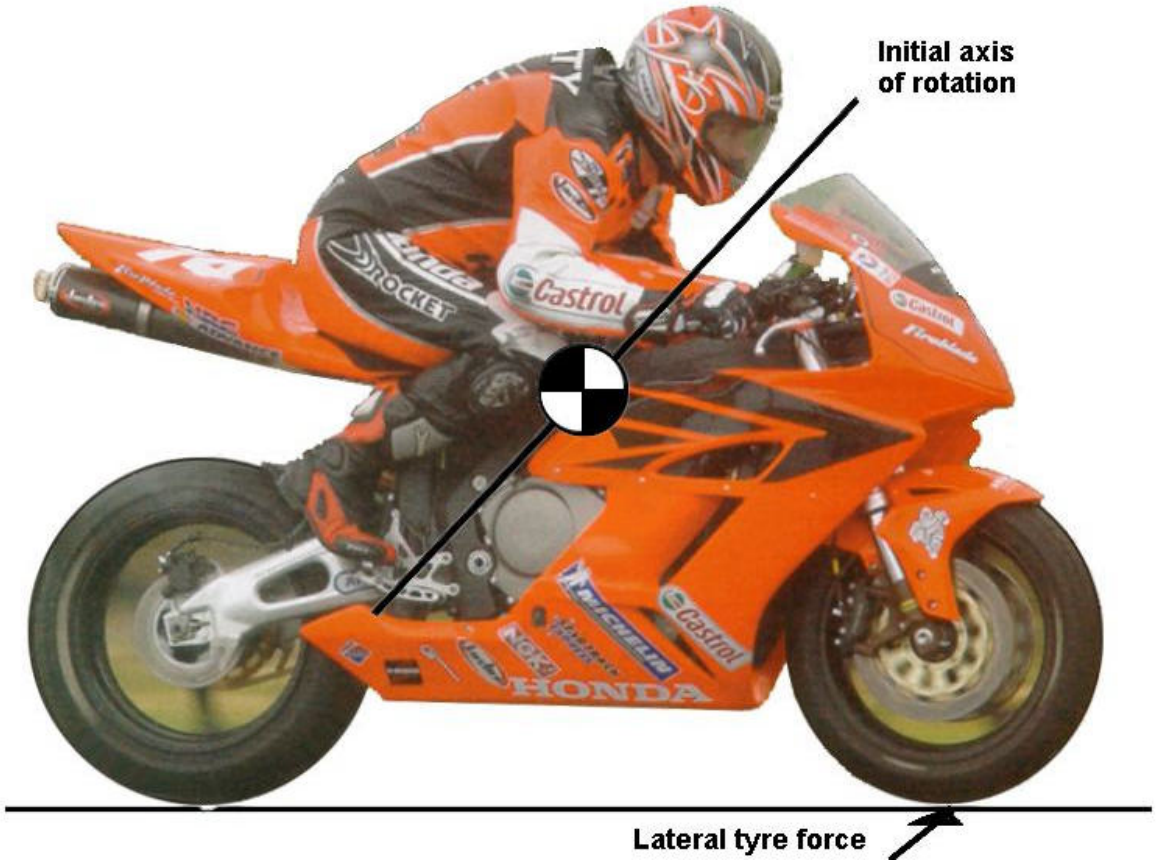

Fig. 4.21 **The counter-steering tyre force is only applied at the front wheel. Initially the rear produces no similar lateral force and so the machine will not rotate purely in roll but will also have a yaw rotation, because the tyre force is offset from both the yaw axis and roll axis through the CoG. These two rotations can be combined into one which will be orientated roughly as shown. The exact angle of the axis will vary with many parameters and as the roll builds up it will become more horizontal.**

The exact angle depends upon the dimensions of the bike and the relationship between the moments of inertia about the three axis through the CoG. The calculation of that angle is further complicated by the tendency of the rear tyre to resist lateral movement, thus keeping the axis close to a line through the rear tyre contact and the CoG. As gravity takes over from counter-steering and begins to provide most of the roll torque this axis will move towards the horizontal, fig 4.22. To achieve a quick lean-in it is thus not

sufficient to just aim for a low roll moment of inertia, a low yaw moment of inertia is also important. This implies a low pitch value too. It is important to try and centralize mass around the loaded CoG, not just around the roll axis through the CoG. The wheels, which of necessity are far from the CoG, are the largest single contributors to roll and yaw inertia. One of the reasons for a short wheelbase and light wheels on racers and sport bikes.

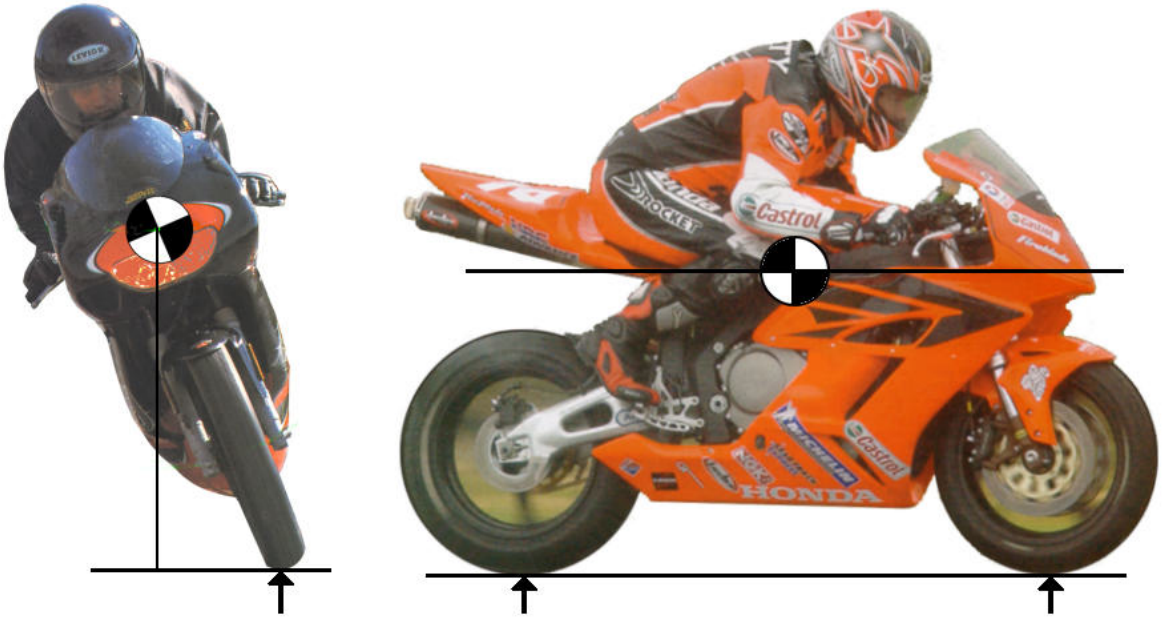

Fig 4.22 **After the initial counter-steer has started the lean process, gravity becomes the predominant source of roll torque and the axis moves from that show in fig 4.17 towards the horizontal.**

In detail

Refer to figs. 4.10 to 4.12. The gyroscopic reactions are of minor importance to the overall roll torque, which for around the first 100 msecs. is due mainly to counter-steering tyre forces. As the counter steering phase comes to an end, a roll angle and velocity will have built up and the gravitational roll torque becomes significant. This gravitational torque is mainly balanced by the opposing tyre induced torque (as the lean angle builds up), except during the counter-steering phase when the gravitational torque is small anyway. Even when the tyre torque and gravitational torque balance, the roll motion continues due to the roll momentum which has been built up. Approaching the final lean angle the rider modifies his control input to ease the roll to its final steady state position.

It is interesting to compare the shapes of the steer/slip angle curve with the applied torque curve. Until the steady state is reached the applied torque always remains negative, however, the steering movement shows a more complex shape. Initially it is negative and this, *(through the tyre steering stiffness)* gives the initial negative tyre lateral force that starts the roll process. During the final part of the manoeuvre, prompted by the reduction in applied torque the slip angle shows a positive bump in its curve, which in turn through the tyre steering stiffness will slow the roll. So if we talk about applied torque then counter-steering lasts for the whole duration of the transient phase, whereas, talking about steering movements the counter-steering only lasts for a brief initial period.

The roll torque is little affected by precession as we have seen but this is far from the case with the steering torque. The roll velocity must be accompanied by a corresponding steering torque. This gyroscopically induced steering torque can easily be a major component of the total steering torque during lean-in. We have seen that these precessional effects are of great importance to the maintenance of auto-stability. Looking at the graphs of steering torques from case 3, we can see that the overall torque available for angular acceleration of the steering is much smaller than any of the individual components, it is the difference between other much larger valued parameters, as such, a small percentage change in any of the contributing torques can have a disproportionably large effect on the overall torque, it is through the negative feedback nature of the gyroscopic torque that this balance is kept under control. Remember how difficult it was to get stable control in the case with no gyroscopic reactions.

Modern racing techniques demand rapid lean-in to corners, so therefore we need to optimise the use of the rider's input as much as possible. From fig. 4.12 we can see that during the initial counter-steering phase the rider's torque is opposed firstly by the tyre force induced steering torque, followed shortly afterward by gyroscopic steering torque. Thus to make better use of the rider's effort we need to minimize the tyre feedback torque and the gyroscopic torque. The tyre feedback is largely dependent on trail and this is one of the prime reasons for trail reduction to speed turn-in, although there are competing influences on the selection of trail. Gyroscopic reaction is dependent on the polar moment of inertia of the wheel and so a reduction in general in wheel weight will also result in improved lean-in.

Performance aspects

Modern racing techniques demand high roll rates, and from the above studies we can elaborate on the most significant physical parameters of importance to achieve the best performance. Within the range of dimensions currently used in racing the most influential is probably the moment of inertia of the wheels, particularly the front. Precessional effects work against rider input and so reducing the required gyroscopic moment will leave more of the rider's effort available to initiate the lean.

During the counter-steering phase, trail works with the lateral tyre force to produce a steering moment into the curve, opposing the rider's input. A low value of trail therefore helps achieve good lean-in performance. Rake angle also affects the roll rate but mainly because of its influence on the trail.

Weight distribution is an important parameter for two quite different reasons. The most obvious is that increased vertical load on the front tyre allows for higher maximum lateral forces, and so the initial counter-steering impulse can be higher. Referring back to fig. 4.21 we see that a forward weight bias, i.e. the CoG is toward the front, makes the initial axis of rotation more horizontal. That means that more of the torque generated about the CoG by the lateral tyre force will be used to start the roll.

Wheelbase is another factor with multiple effects. One is simply the Ackerman type steering considerations as shown in fig. 3.20. The build up of a counter-steer yaw will be reduced and slower, with a longer wheelbase, which means that the rear tyre will provide a reduced contribution to roll torque. Increased yaw inertia is probably a more important parameter, which will slow the turn initialisation with longer machines. There is a countering tendency which partially compensates for the slowing effects of a long wheelbase, the initial rotation axis, as in fig. 4.21, will be more horizontal as in the forward weight distribution case above.

Moments of inertia, principally roll and yaw, have a direct affect on roll and turn rates. Angular acceleration is inversely proportional to moments of inertia for a given turn inputs. We are interested in both roll and yaw accelerations. Initial roll from counter-steering is needed to start the motorcycle lean, before gravity takes over. Yaw response is necessary to recover as quickly as possible from a counter-yaw attitude from the counter-steering, and then build up the correct yaw velocity around the curve.